

Project Design Phase

Solution Architecture

Date: 26 June 2026

Team ID: XXXXX

Project Name: Smart Lender

Maximum Marks: 4 Marks

Solution Architecture

The **Smart Lender** system is a Machine Learning-based web application that predicts whether a loan application is likely to be approved or rejected. The architecture integrates a user-friendly web interface, a Flask backend, and a trained **Random Forest (rdf.pkl)** model to automate the loan approval prediction process.

The application follows a layered architecture where user inputs are validated, processed, analysed by the Machine Learning model, and the prediction result is displayed instantly.

System Workflow

1. The user opens the Smart Lender application.
2. The user enters personal and financial details.
3. The frontend sends the information to the Flask backend.
4. Flask validates and preprocesses the input data.
5. The processed data is passed to the trained **Random Forest (rdf.pkl)** model.
6. The model predicts the loan approval status.
7. Flask receives the prediction and sends the result back to the frontend.
8. The application displays **Loan Approved** or **Loan Rejected** instantly.

Technologies Used

Component	Technology
Frontend	HTML5, CSS3
Backend	Python, Flask
Machine Learning	Random Forest Classifier (Scikit-learn)
Data Processing	Pandas, NumPy
Model Storage	Pickle (rdf.pkl)
Development Environment	Visual Studio Code

Benefits of the Architecture

- Modular and easy to maintain.
- Fast and accurate loan predictions.
- Simple user interface for applicants.
- Scalable for future enhancements such as database integration, user authentication, and cloud deployment.
- Reduces manual effort and improves decision consistency.

Conclusion

The **Smart Lender Solution Architecture** provides a clear workflow connecting the frontend, backend, Machine Learning model, and data layer. This design enables fast, reliable, and data-driven loan approval predictions while ensuring that the application is scalable, maintainable, and suitable for real-world banking and financial services.